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ORIGINAL ARTICLE

Alterations in Oral Microbiota in HIV Are Related to Decreased Pulmonary Function

Libing Yang[1,2,3], Daniel G. Dunlap[1], Shulin Qin[1,2], Adam Fitch[2], Kelvin Li[2], Carl D. Koch[1,2], Mehdi Nouraie[1], Rebecca DeSensi[1], Ken S. Ho[4], Jeremy J. Martinson[5], Barbara Methé[1,2], and Alison Morris[1,2,6]

[1]Division of Pulmonary, Allergy and Critical Care Medicine and [2]Center for Medicine and the Microbiome, Department of Medicine, [4]Division of Infectious Disease, Department of Medicine, and [6]Department of Immunology, University of Pittsburgh School of Medicine, Pittsburgh, Pennsylvania; [3]School of Medicine, Tsinghua University, Beijing, China; and [5]Infectious Diseases and Microbiology, University of Pittsburgh School of Public Health, Pittsburgh, Pennsylvania

ORCID ID: 0000-0003-2564-207X (C.D.K.).

Abstract

Rationale: Mechanisms of HIV-associated chronic obstructive pulmonary disease (COPD) are poorly understood. The oral microbiome shapes the lung microbiome, and gut dysbiosis can affect lung diseases; however, relationships of the oral and gut microbiome to COPD in HIV have not been explored.

Objectives: To examine alterations in the oral and gut microbiome associated with pulmonary disease in persons with HIV (PWH).

Methods: Seventy-five PWH and 93 HIV-uninfected men from the MACS (Multicenter AIDS Cohort Study) performed pulmonary function testing. Sequencing of bacterial 16S ribosomal RNA in saliva and stool was performed. We used nonmetric multidimensional scaling, permutational multivariate ANOVA, and linear discriminant analysis to analyze communities by HIV and lung function.

Measurements and Main Results: Oral microbiome composition differed by HIV and smoking status. Alterations of oral microbial communities were observed in PWH with abnormal lung function with increases in relative abundance of *Veillonella*, *Streptococcus*, and *Lactobacillus*. There were no significant associations between the oral microbiome and lung function in HIV-uninfected individuals. No associations with HIV status or lung function were seen with the gut microbiome.

Conclusions: Alterations of oral microbiota in PWH were related to impaired pulmonary function and to systemic inflammation. These results suggest that the oral microbiome may serve as a biomarker of lung function in HIV and that its disruption may contribute to COPD pathogenesis.

Keywords: HIV; microbiome; saliva; stool; lung function

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Correspondence and requests for reprints should be addressed to Alison Morris, M.D., M.S., UPMC Montefiore Hospital, NW628, 3459 Fifth Avenue, Pittsburgh, PA 15213. E-mail: morrisa@upmc.edu.

This article has a related editorial.

This article has an online supplement, which is accessible from this issue's table of contents at www.atsjournals.org.

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At a Glance Commentary

Scientific Knowledge on the Subject: Antiretroviral therapy has improved survival in persons with HIV, but chronic obstructive pulmonary disease (COPD) still causes morbidity and mortality. HIV infection may be an independent risk factor for COPD; however, the underlying mechanisms are poorly understood. The lung is exposed to the oral microbiome via microaspiration, and changes in the oral microbiome may affect lung disease. The gut microbiome can have distant effects on other end organs by impact on the immune response and inflammation, by translocation, or through metabolic products.

What This Study Adds to the Field: We performed bacterial 16S ribosomal RNA sequencing of saliva and stool in the Pittsburgh site of the MACS (Multicenter AIDS Cohort Study) of persons with HIV and HIV-uninfected individuals, with and without COPD. We identified oral microbiome differences by HIV and smoking status. We discovered a novel association of oral microbiome with lung function measured by post-bronchodilator FEV_1/FVC or reduced D_{LCO}. There were no significant relationships of stool microbiome to lung function. This study suggests that the oral microbiome may play an important role in HIV COPD pathogenesis, possibly by stimulating inflammation and promoting lung damage.

Antiretroviral therapy (ART) has improved survival in persons with HIV (PWH), but lung disease, including chronic obstructive pulmonary disease (COPD), remains a cause of morbidity and mortality (1–4). The underlying mechanisms involved in COPD pathogenesis are poorly understood. Lifestyle choices such as increased cigarette smoking contribute to disease but cannot fully explain of the pathogenesis of COPD in PWH (2, 5–7).

The composition and function of resident bacteria can have a profound impact on the immune system, inflammation, and metabolite production that may lead to disease or exacerbate preexisting conditions (8). Lung bacterial communities largely resemble those in the mouth because of subclinical microaspiration and dispersion along bronchial mucosa (9–11), and alterations of oral bacteria have been implicated in COPD in the HIV-uninfected population (12). The Lung HIV Microbiome Project demonstrated alterations in the oral bacterial microbiome in HIV and in smokers (9, 13–17). Given the contribution of oral microbes to the lung environment, it is possible that changes in the oral microbiome could stimulate inflammation that contributes to lung disease, particularly in smokers or PWH. Prior work has shown that healthy individuals with oral taxa predominance in BAL have increased inflammatory mediators (18), but no studies have evaluated the relationship between oral bacterial microbiota and COPD in PWH. In addition, oral samples are easily accessible, and modification of the oral microbiome could indirectly affect lung bacteria, making examination of oral samples useful for investigating and potentially modifying the lung microbiome in disease.

Recent evidence shows that gut microbiome and its metabolic products in PWH differ from those in HIV-uninfected individuals and that these alterations correlate with immune defects and chronic inflammation (19–21). Dysbiosis in gut microbes may have an impact on lung disease through effects on systemic inflammation, microbial translocation, or production of various metabolites (22–26), but their relationships have not been evaluated in COPD associated with HIV.

We investigated the relationship between oral and gut microbiome and lung function in PWH and HIV-uninfected individuals. We collected saliva and stool from PWH and HIV-uninfected individuals with or without COPD and performed next-generation sequencing of the bacterial 16S ribosomal RNA gene. Some results of these studies were reported previously in the form of abstracts (27–30).

Methods

Study Participants

We recruited 168 participants (75 PWH and 93 HIV-uninfected individuals) from the Pittsburgh site of the MACS (Multicenter AIDS Cohort Study). The MACS is a longitudinal cohort study of men with or at risk for HIV. All participants without a contraindication to pulmonary function testing were eligible. Exclusion criteria included individuals with active respiratory infection, those receiving antibiotic treatment in the previous 5 days, and individuals receiving systemic steroid therapy in the previous 5 months. We also performed analyses excluding individuals who had used antibiotics within 6 months. Because results were not significantly different, we used the 5-day exclusion to maintain sample size. Demographic, physical, and laboratory variables were collected during the regular MACS visit. All participants provided written informed consent. The University of Pittsburgh institutional review board approved this study, and detailed information regarding recruitment is available in the online supplement.

Pulmonary Function Testing

Participants performed pre- and post-bronchodilator spirometry using 360 μg of albuterol, per American Thoracic Society (ATS) guidelines (31, 32). The third National Health and Nutrition Examination Survey equations were used for spirometry reference values (31). D_{LCO} was measured following ATS and European Respiratory Society standards (33).

D_{LCO} was adjusted for Hb and carboxyhemoglobin, and reference values for D_{LCO} were calculated using Neas equations (34). All pulmonary function tests (PFTs) were performed by trained research technicians, and acceptable quality was confirmed by a central reading center, as per ATS guidelines (32).

Sample Collection

Saliva samples were collected at time of PFT. Participants refrained from eating, drinking, smoking, or chewing gum for 30 minutes before saliva collection. Unstimulated saliva samples were collected in DNA/RNA stabilizing buffer (DNA/RNA Shield; ZYMO Research). Stool samples were collected at home in the stabilizing buffer within a week of PFT. All samples were aliquoted and frozen at −80°C within 30 days of collection.

Sample Processing and Sequencing

We extracted bacterial DNA directly from saliva and stool samples using DNeasy PowerSoil Kits (Qiagen). We amplified the bacterial 16S ribosomal RNA gene V4 subunit using PCR (35, 36), and we performed sequencing on the Illumina MiSeq platform (35). Negative controls were used per batch of DNA extraction and PCR to examine possible contamination (Figure E1 in the online supplement).

Sequence Processing

We demultiplexed reads onboard the sequencer using standard Illumina software and performed postsequencing quality control—including sequence filtering and trimming on raw 16S sequences from samples and negative controls—using an in-house pipeline developed by the University of Pittsburgh Center for Medicine and the Microbiome, utilizing open source software including *dust*, *fastx toolkit*, and *cutadapt* (37–39). We applied an in-house *mothur* pipeline to generate operational taxonomic units and ribosomal database project classified sequences (40, 41).

Plasma Biomarker Detection

Plasma levels of 6 biomarkers were measured with a Luminex assay (Bio-Rad), including IL-8, GCSF (granulocyte colony–stimulating factor), IFN-γ, MCP1 (monocyte chemoattractant protein-1), MIP-1β (macrophage inflammatory protein-1β), and TNF-α (tumor necrosis factor α). Plasma levels of IL-6, sCD163, and endothelin-1 were measured with ELISA (R&D Systems). These biomarkers were selected as having previous relationships with HIV-associated COPD (42–48).

Statistical Analysis

We compared participants' characteristics between PWH and HIV-uninfected individuals using the *t* test, chi-square test, Mood median test, and Fisher exact test, as appropriate. All participants were stratified by HIV and smoking status (current smokers, former smokers, and nonsmokers). We selected $FEV_1/FVC < 0.70$ as the Global Initiative for Chronic Obstructive Lung Disease definition for airway obstruction and D_{LCO} <80% predicted as the clinical definition of at least mild impairment. We also used the lower limit of normal as a cutoff and compared the results generated by each definition.

We performed ecological analysis of α diversity (by Shannon index) with the R *vegan* package (49). We applied the nonmetric multidimensional scaling plot using the Bray–Curtis dissimilarity index to visualize clustering of samples. Univariate permutational multivariate ANOVA (PERMANOVA) testing was performed to compare the microbiome between different groups with the R *vegan* package. Multivariate PERMANOVA controlling for age, race, smoking status, CD4 count, and HIV viral load was performed using a custom pipeline based on previous studies indicating that these variables could influence microbiome compositions (50–53). Lung function parameters were used as continuous variables in PERMANOVA. Linear discriminant analysis effect size was performed on Galaxy to identify genera that were differentially abundant between groups (54). We set α values for the factorial Kruskal–Wallis at 0.05 and the linear discriminant analysis score threshold for discriminative features at 2.0.

Table 1. Characteristics of Participants

	Persons with HIV	HIV-uninfected Individuals	*P* Value
n	75	93	—
Age, yr, mean (SD)	55.7 (11.2)	57.5 (12.3)	0.34*
Race, *n* (%)			
White	57 (76.0)	72 (77.4)	0.30†
Black	18 (24.0)	18 (19.4)	
Hispanic, *n* (%)	0 (0)	3 (3.2)	0.25†
Smoking status, *n* (%)			
Nonsmoker	26 (34.7)	31 (33.3)	0.28‡
Current smoker	23 (30.7)	20 (21.5)	
Former smoker	26 (34.7)	42 (45.2)	
ART, *n* (%)	72 (96.0)	NA	—
Detectable HIV RNA viral load, *n* (%)	3 (4.0)	NA	—
Current CD4 count, cells/μl, median (range)	743 (187–1,422)	925 (267–2,287)	0.005§
Post-bronchodilator FEV_1/FVC, mean (SD)	0.78 (0.09)	0.77 (0.09)	0.73*
Post-bronchodilator FEV_1/FVC <0.7, *n* (%)	12 (16.0)	10 (10.8)	0.43‡
Post-bronchodilator FEV_1% predicted, mean (SD)	0.96 (0.17)	0.95 (0.16)	0.94*
Post-bronchodilator FEV_1% predicted <80%, *n* (%)	6 (8.0)	12 (12.9)	0.45‡
D_{LCO} % predicted, mean (SD)	0.81 (0.16)	0.86 (0.14)	0.09*
D_{LCO} % predicted <80%, *n* (%)	29 (38.7)	29 (31.2)	0.38‡
D_{LCO} % predicted <60%, *n* (%)	8 (10.7)	2 (2.2)	0.02†

Definition of abbreviations: ART = antiretroviral therapy; NA = not available.
Participant characteristics were summarized using means, percentages, or medians and compared using the *t* test, chi-square test, Fisher exact test, or Mood median test, as appropriate.
**t* test.
†Fisher exact test.
‡Chi-square test.
§Mood median test.

All biomarker levels were normalized. Abundances of genera were transformed with the additive log ratio transform to mitigate spurious correlation among taxa due to the compositional nature of the microbiota profiles. The transformation permits each taxonomic abundance to be treated as independent (from each other) and normally distributed (55). Multiple linear regression was used to examine associations of biomarkers with the additive log ratio of signature genera and lung function parameters, correcting for age, race, and HIV viral load.

Results

Participants

Seventy-five PWH and 93 HIV-uninfected participants were included. PWH and HIV-uninfected individuals were similar in age, race, ethnicity, and smoking status (Table 1). In total, 162 participants (96%) had acceptable spirometry per ATS/European Respiratory Society criteria, and 156 (93%) participants had acceptable D_{LCO} testing. Participants who had acceptable PFTs were not significantly different from those who did not in terms of age, race, ethnicity, and smoking status.

Oral Microbiome Analyses

Overall, 160 saliva samples were available for analysis. The most dominant genera detected within the oral microbiome were *Prevotella*, *Veillonella*, *Neisseria*, Pasteurellaceae (family level, genus not classified), *Fusobacterium*, *Rothia*, and *Streptococcus*, which accounted for 80% of the overall abundance of bacteria in the mouth.

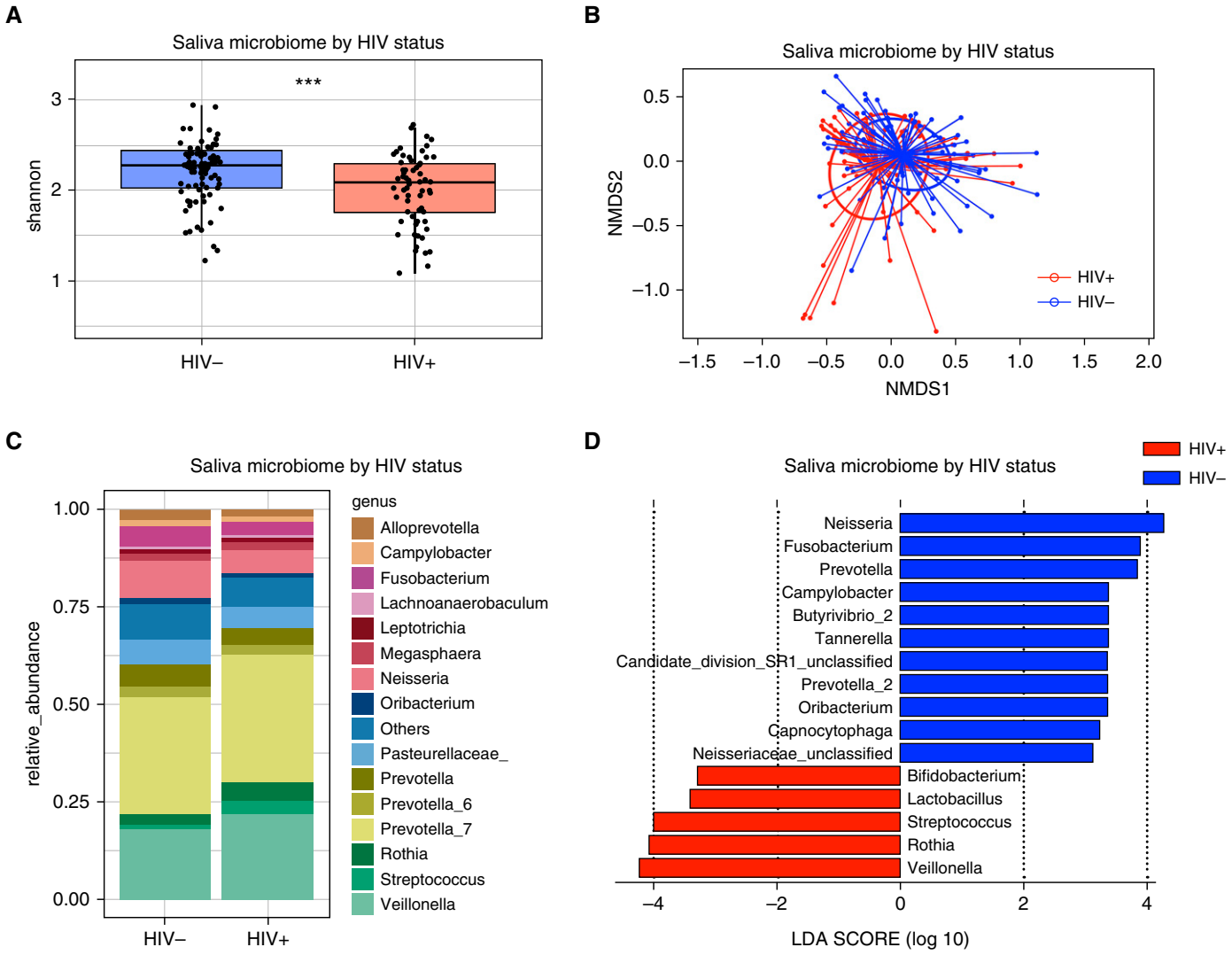


Figure 1. Oral microbiome comparisons between persons with HIV (PWH) and HIV-uninfected individuals. (*A*) PWH had significantly lower oral microbiome α diversity than HIV-uninfected individuals (by Shannon index, $P = 0.00083$). (*B*) The β diversity of the oral microbiome differed in PWH compared with HIV-uninfected individuals (permutational multivariate ANOVA for Bray–Curtis distance, $P < 0.01$). (*C*) Ranked abundances of the genera present in the mouth are depicted with stacked bar plots. (*D*) Linear discriminant analysis effect size demonstrates genera that were significantly increased in individuals with PWH (red) or HIV-uninfected individuals (blue) in saliva. ***$P < 0.005$. LDA = linear discriminant analysis; NMDS = nonmetric multidimensional scaling.

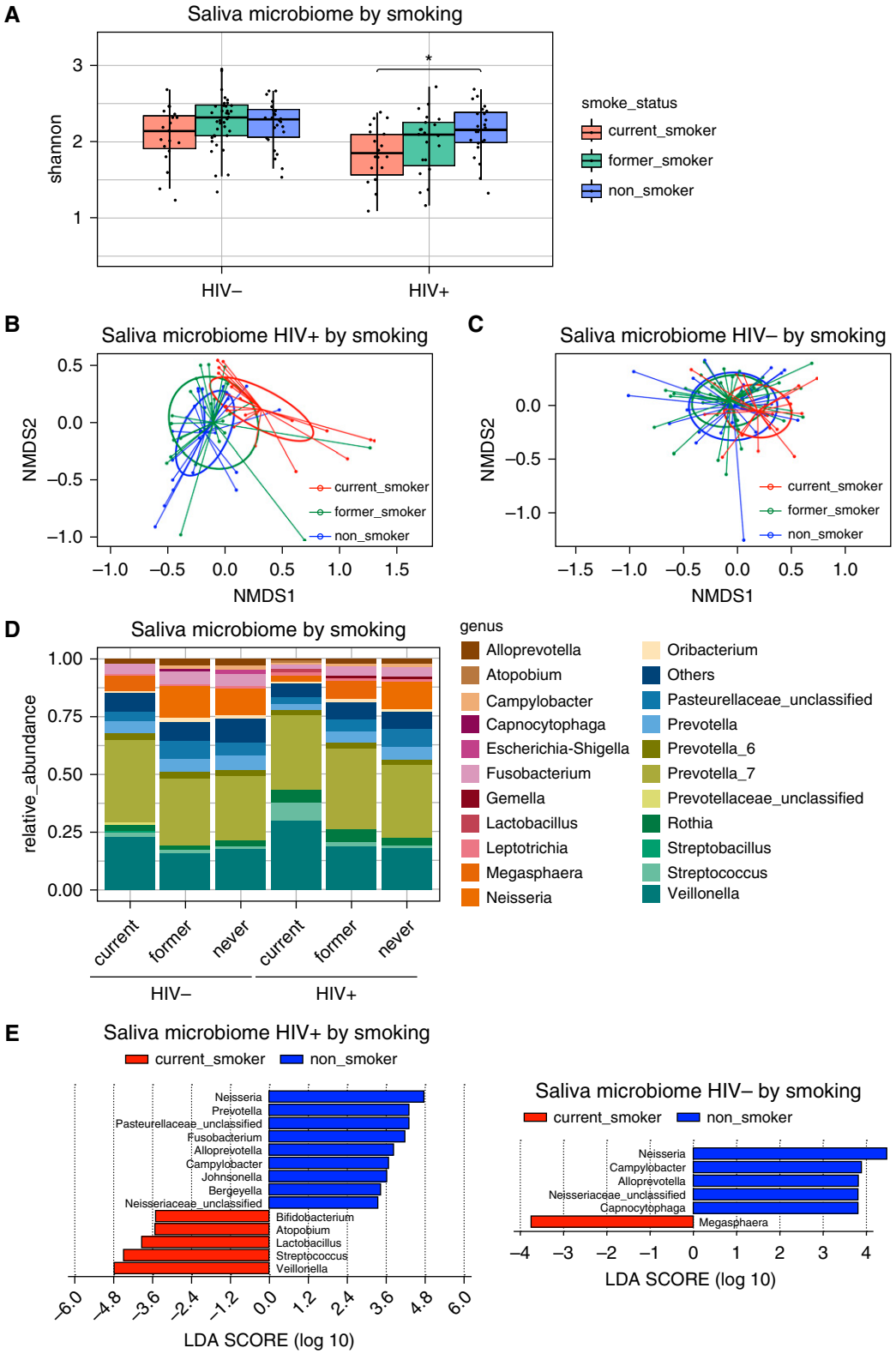


Figure 2. Oral microbiome comparisons among current smokers, former smokers, and nonsmokers by HIV status. (*A*) HIV-infected current smokers had reduced oral microbiome α diversity (by Shannon index, $P = 0.013$). (*B* and *C*) Smoking status was associated with β diversity of the oral microbiome in persons with HIV (*B*) and HIV-uninfected individuals (*C*) (permutational multivariate ANOVA for Bray–Curtis distance, $P < 0.05$). (*D*) The most abundant bacteria are depicted with stacked bar plots, grouped by HIV and smoking status. (*E*) Differential bacterial abundance in oral samples by smoking status is depicted using linear discriminant analysis effect size. $^*P < 0.05$. LDA = linear discriminant analysis; NMDS = nonmetric multidimensional scaling.

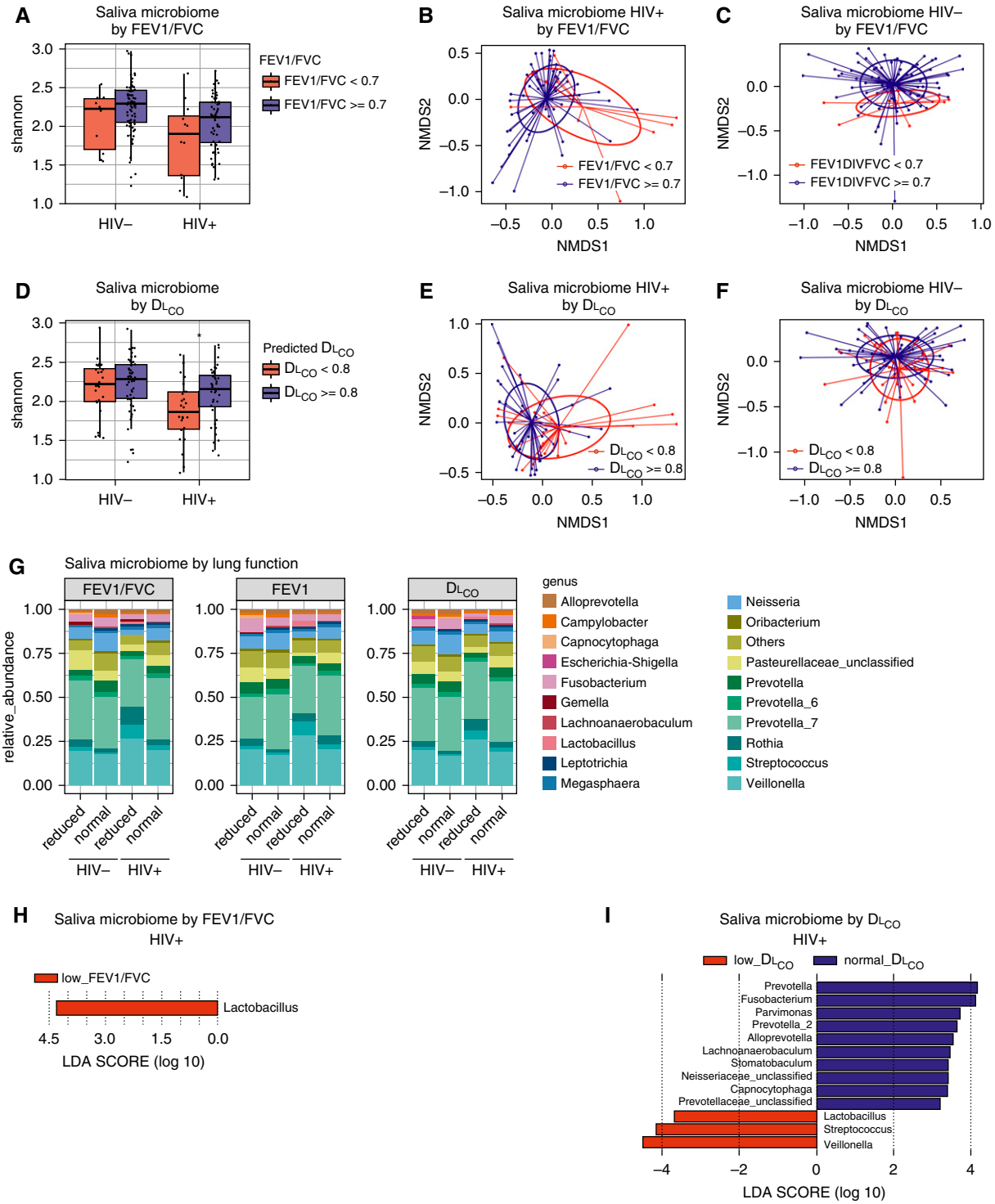


Figure 3. Oral microbiome comparisons between participants with and without lung dysfunction by HIV status. (*A*) Participants with and without reduced FEV_1/FVC had similar α diversity of oral microbiome regardless of HIV status. (*B* and *C*) NMDS and permutational multivariate ANOVA (PERMANOVA) demonstrated oral microbiome alteration in persons with HIV (PWH) versus without reduced FEV_1/FVC (PERMANOVA for Bray–Curtis distance, $P < 0.05$) (*B*) but not in HIV-uninfected individuals (*C*). (*D*) The α diversity of oral microbiome of PWH with reduced D_{LCO} was lower compared with those without reduced D_{LCO} (by Shannon index, $P = 0.033$). (*E* and *F*) There were significant differences in oral microbiome composition between individuals with and without reduced D_{LCO} among PWH (PERMANOVA for Bray-Curtis distance, $P < 0.05$) (*E*) but not among HIV-uninfected individuals (*F*). (*G*) The most abundant bacteria are depicted with stacked bar plots, grouped by HIV and lung function parameters. (*H* and *I*) Differential bacterial abundance in oral samples from PWH by FEV_1/FVC (*H*) and D_{LCO} (*I*) is depicted using linear discriminant analysis effect size. *$P < 0.05$. LDA = linear discriminant analysis; NMDS = nonmetric multidimensional scaling.

Comparison of PWH and HIV-uninfected Individuals

The α diversity in saliva from PWH was significantly reduced compared with HIV-uninfected individuals (*t* test, $P = 0.00083$; Figure 1A). The β diversity was also different between groups in both univariate analysis (PERMANOVA $P = 0.01$, $R^2 = 0.016$) and multivariate analyses controlling for age, race, and smoking status (PERMANOVA $P = 0.0006$, $R^2 = 0.028$; Figure 1B). Among PWH, neither α nor β diversity differed by ART, CD4 count, or HIV viral load (data not shown). We visualized the relative abundance of dominant taxa at the genus level in the oral microbiome of individuals by HIV status (Figure 1C). Abundance of *Veillonella*, *Rothia*, and *Streptococcus* were significantly increased in the oral microbiome of PWH, whereas abundance of *Neisseria* was significantly decreased (Figure 1D).

Comparison by Smoking Status

We compared α diversity of the oral microbiome by smoking status. Among PWH, those who were actively smoking had reduced α diversity in saliva compared with nonsmokers (*t* test, $P = 0.013$; Figure 2A). In HIV-uninfected individuals, oral α diversity of current smokers was higher than that of former smokers, although it was not significant (*t* test, $P = 0.06$; Figure 2A). In both univariate and multivariate analyses controlling for age, race, CD4 count, and HIV RNA viral load, there was strong difference in overall oral microbiome community structure among current smokers, former smokers, and nonsmokers in PWH (in univariate analysis, PERMANOVA $P = 0.017$, $R^2 = 0.034$; in multivariate analysis, PERMANOVA $P = 0.0004$, $R^2 = 0.12$; Figure 2B). In HIV-uninfected individuals, we observed a similar relationship between the oral microbiome and smoking status using both models (in univariate analysis, PERMANOVA $P = 0.022$, $R^2 = 0.0323$; in multivariate analysis, PERMANOVA $P = 0.011$, $R^2 = 0.052$; Figure 2C). Relative abundance of the top 15 dominant taxa pointed toward shared and unique taxa of oral microbiome (Figure 2D) in each group. Oral microbes such as *Veillonella*, *Streptococcus*, and *Lactobacillus* were more abundant in PWH who were current smokers compared with nonsmokers, whereas *Neisseria*, *Prevotella*, Pasteurellaceae (family level, genus not classified), and *Fusobacterium* were relatively decreased. HIV-uninfected current smokers had less *Neisseria* in the oral microbiome (Figure 2E).

Comparison by Lung Function

We evaluated the relationship between the oral microbiome and lung function in PWH and HIV-uninfected individuals. There was no significant difference in α diversity of the oral microbiome between groups with or without reduced post-bronchodilator FEV_1/FVC regardless of HIV status (*t* test, not significant; Figure 3A). The β diversity was different among PWH with airway obstruction, as measured by reduced post-bronchodilator FEV_1/FVC compared with those without (in univariate analysis, PERMANOVA $P = 0.0281$, $R^2 = 0.033$; in multivariate analysis, PERMANOVA $P = 0.036$, $R^2 = 0.037$; Figure 3B). In HIV-uninfected individuals, no differences were seen by FEV_1/FVC using multivariate analysis controlling for age, race, and smoking status (Figure 3C). There was no association of FEV_1 and oral microbiome composition in either α or β diversity regardless of HIV status (data not shown).

Among PWH, those with impaired D_{LCO} had significantly lower oral α diversity (*t* test, $P = 0.033$; Figure 3D). The β diversity also differed in these individuals compared with PWH with a normal predicted D_{LCO}, using both univariate and multivariate models (in univariate analysis, PERMANOVA $P = 0.0051$, $R^2 = 0.046$; in multivariate analysis, PERMANOVA $P = 0.015$, $R^2 = 0.043$; Figure 3E). In HIV-uninfected individuals, D_{LCO} was not related to α or β diversity (Figures 3D and 3F).

We visualized the relative abundance of dominant taxa (top 15) at the genus level in the oral microbiome of individuals by HIV status and lung function with stacked bar plots (Figure 3G). In oral samples, increases in the relative abundance of *Veillonella*, *Streptococcus*, and *Lactobacillus* were associated with abnormal lung function in PWH (Figures 3H and 3I). Similar results were observed when using the lower limit of normal as the cutoff for airway obstruction (Figure E2).

Associations of Inflammatory Biomarkers, PFTs, and Microbiome in PWH

In PWH, plasma TNF-α, endothelin-1, and MIP-1β were negatively correlated with lung function parameters (Figure 4). We then examined the associations of these three plasma biomarkers to key genera. Plasma TNF-α and MIP-1β were increased in PWH with more abundant oral *Streptococcus* and *Lactobacillus.* Plasma endothelin-1 levels were also positively correlated with oral *Streptococcus*, *Veillonella*, and *Lactobacillus* abundance (Figure 5).

Gut Microbiome and Lung Function

Gut microbiome analyses are available in the online supplement.

Discussion

We investigated oral bacterial microbiota alterations in PWH and COPD. We observed oral microbiome differences by HIV and smoking status. Importantly, we found significant differences in the oral bacterial community that were associated with lung function abnormalities in PWH and correlation of plasma inflammatory biomarkers with lung dysfunction and oral taxonomic signature. These findings suggest that changes in the oral microbiome may be involved in the pathogenesis of lung disease in PWH, possibly by stimulating inflammation, and that the oral microbiome may serve as an easily accessible marker of lung dysfunction.

As in previous work, the oral microbiome of PWH differed from that of HIV-uninfected individuals (13, 15, 51). PWH had decreased oral microbiome α diversity, and there were differences in β diversity compared with HIV-uninfected controls. The Lung HIV Microbiome Project compared the oral and lung microbiome of PWH with HIV-uninfected individuals and reported increased abundance of oral *Streptococcus*, *Actinomyces*, and *Rothia* in PWH (14). Our results were similar in identifying increases in oral *Veillonella*, *Rothia*, and *Streptococcus* and decreased *Neisseria* in PWH. Prior work also reported decreased oral microbial α diversity in PWH with low CD4 counts after 24 weeks of ART (52). We did not find similar differences, possibly because most PWH in our cohort were on long-term ART and had normal CD4 cell counts. Mechanisms explaining compositional

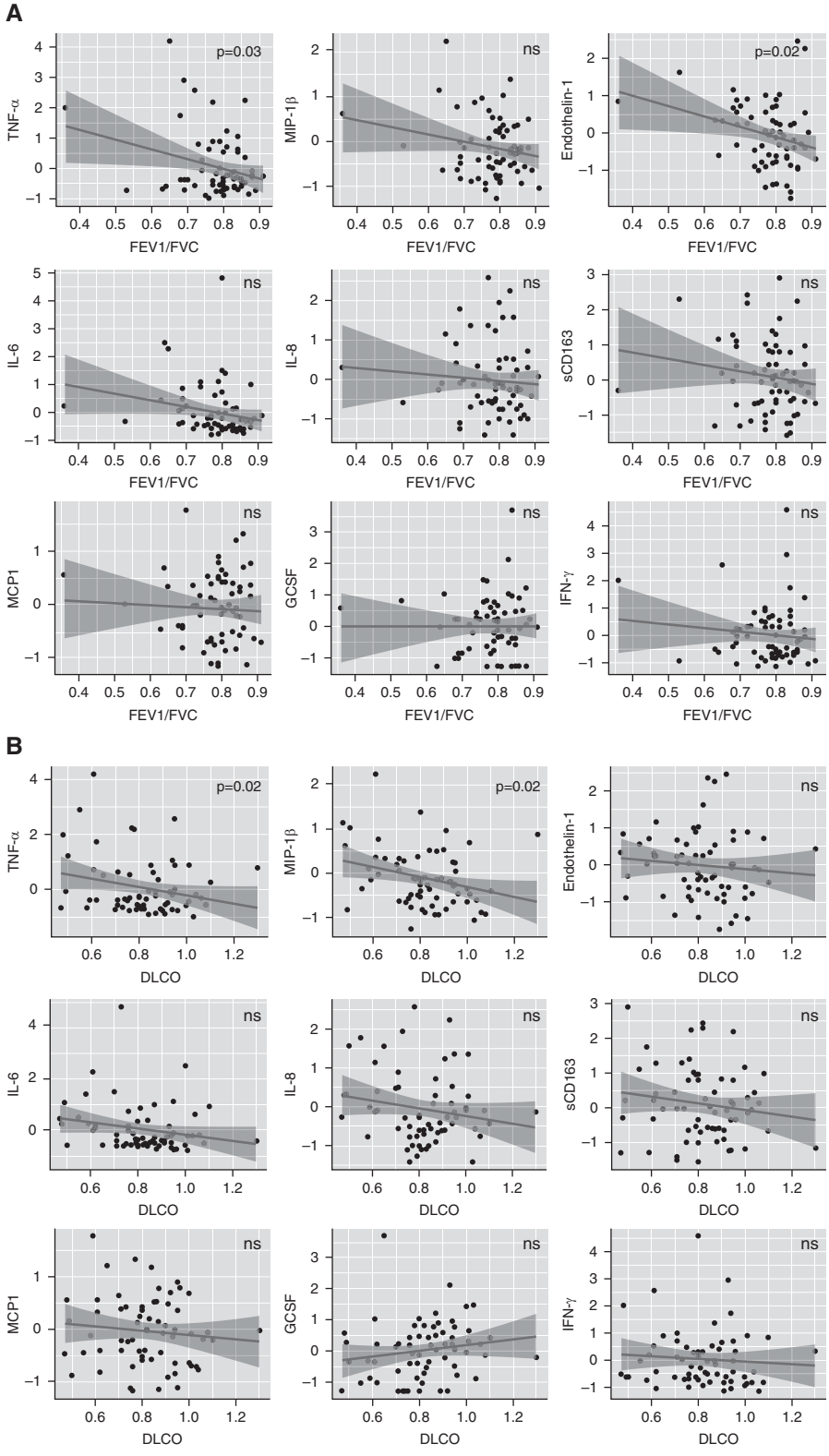


Figure 4. Associations of inflammatory biomarkers with pulmonary function in persons with HIV (PWH). (*A*) Plasma TNF-α and endothelin-1 were negatively associated with FEV_1/FVC in PWH. (*B*) Plasma TNF-α and MIP-1β were negatively associated with predicted D_{LCO} in PWH. GCSF = granulocyte colony–stimulating factor; MCP1 = monocyte chemoattractant protein-1; MIP-1β = macrophage inflammatory protein-1β; ns = not significant; TNF-α = tumor necrosis factor α.

differences in the oral microbiome between PWH and HIV-uninfected are not known but could reflect the aberrant immune activation seen in PWH (56), leading to an altered host response to oral bacteria. Other factors such as dental health, oral hygiene, ART, or diet could also have an impact.

Smoking was also associated with alterations in oral microbiome communities. Current smokers had reduced oral microbiome α diversity compared with nonsmokers only among PWH. And we observed overall compositional alterations in the oral microbiome when comparing current smokers with former smokers and nonsmokers among both PWH and HIV-uninfected individuals. It has been proposed that smoking promotes bacterial colonization of the upper airways by facilitating epithelial binding and impairing mucociliary clearance in healthy participants (57, 58). Our study suggested that the effects of smoking may be even more pronounced in PWH.

Changes in the oral microbiome may be important in lung function. In the HIV-uninfected population, oral health and oral bacteria have been postulated to play roles in COPD (12, 59). Prior work has found alterations in oral health associated with COPD symptoms and severity, presumably because of changes in oral bacteria (60), although one study demonstrated no overall structural differences in periodontal plaque microbiome between individuals with and without COPD (61). Studies of periodontal interventions to improve COPD have shown some improvement in lung function but have not directly assessed changes in the oral microbiome (62).

We discovered significant differences in the oral microbiome related to pulmonary function in PWH. Among PWH, alterations in the oral microbiome β diversity were strongly associated with both FEV_1/FVC and D_{LCO}, and richness of the oral microbiome was reduced in PWH who had impaired D_{LCO}. We identified increased *Veillonella*, *Streptococcus*, and *Lactobacillus* in PWH with abnormal lung function. Certain species of *Veillonella* and *Streptococcus*, such as *Veillonella parvula*, have the ability to induce the production proinflammatory cytokines, including IL-8, IL-6, IL-10, and TNF-α (63, 64). A strong correlation was established between saliva *Lactobacillus* counts and poor dental health, with higher caries correlating with higher *Lactobacillus* counts (65, 66).

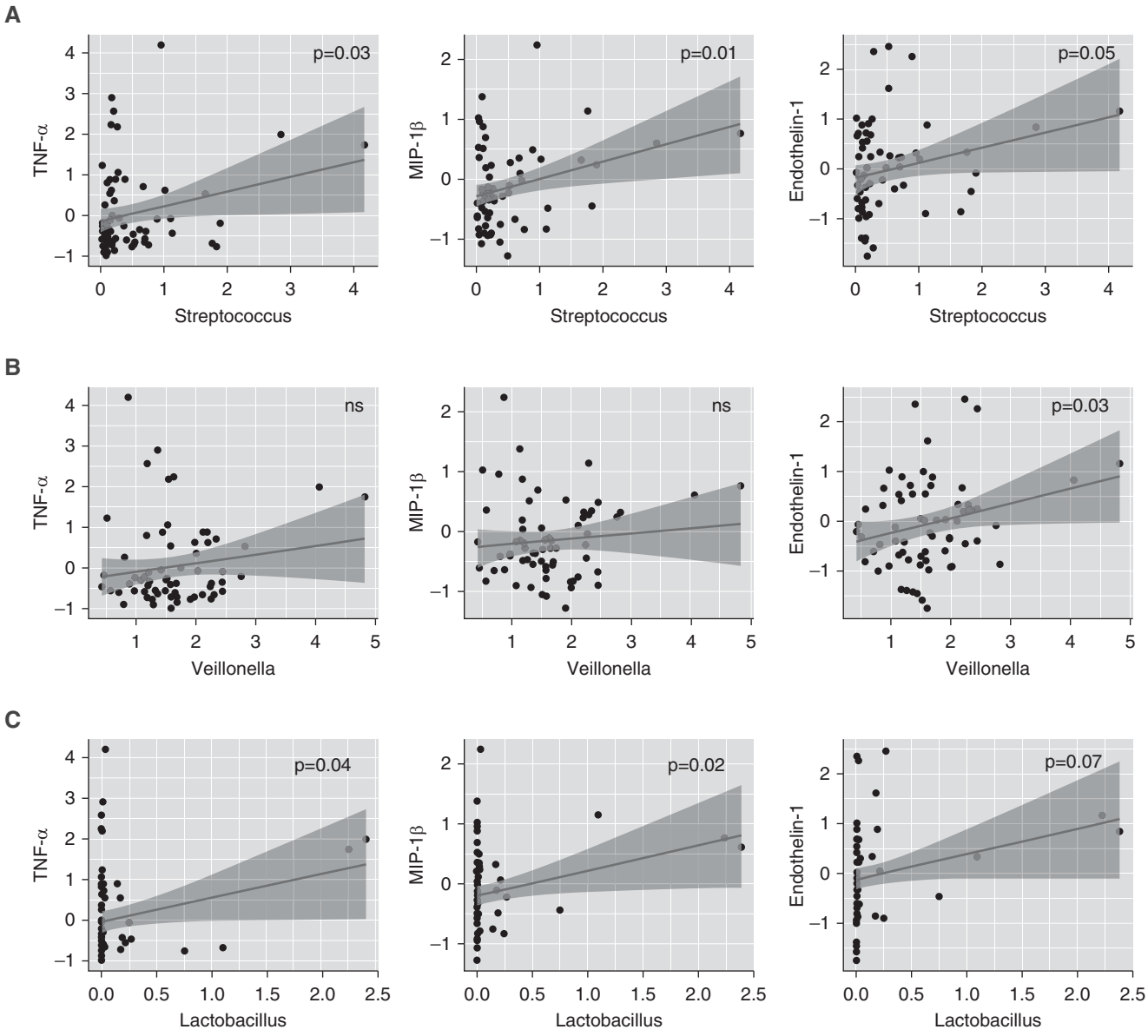


Figure 5. Associations of inflammatory biomarkers to key bacteria in persons with HIV (PWH). (*A*) Plasma TNF-α, MIP-1β, and endothelin-1 were increased in PWH with more abundant oral *Streptococcus*. (*B*) Plasma endothelin-1 was increased in PWH with more abundant oral *Veillonella*. (*C*) Plasma TNF-α, MIP-1β, and endothelin-1 were increased in PWH with more abundant oral *Lactobacillus.* MIP-1β = macrophage inflammatory protein-1β; ns = not significant; TNF-α = tumor necrosis factor α.

Furthermore, a study indicated that some *Lactobacillus* strains could facilitate inflammatory responses of macrophages by inducing proinflammatory cytokines such as IL-8, TNF-α, IL-12p70, and IL-6 (67), although certain species and strains of *Lactobacillus*, such as *Lactobacillus rhamnosus*, can have an antiinflammatory effect (68, 69). Metagenomic sequencing of microbial community with a resolution at strain level is needed in the future. In our study, we observed increases of plasma TNF-α, MIP-1β, and endothelin-1 level associated with decreased lung function; in addition, the levels of these COPD-associated proinflammatory cytokines were positively correlated with the abundance of key bacteria associated with lung dysfunction—*Veillonella*, *Streptococcus*, and *Lactobacillus*—suggesting that the oral microbiome in PWH may contribute to lung dysfunction possibly by simulating inflammatory responses. When activated, macrophages express inflammatory cytokines including TNF-α and MIP-1β, which recruit and activate lymphocytes, leading to pulmonary damage (70, 71). Endothelin-1, a vasoconstrictor and a marker of vascular injury, is also released from activated macrophages. Endothelin-1 has a role in pathogenesis of pulmonary vasculopathy, and that may contribute to impaired diffusing capacity, particularly in PWH (42, 72). There may be different phenotypes of lung function abnormalities such as airway obstruction or impaired diffusing capacity in HIV COPD. D_{LCO} in

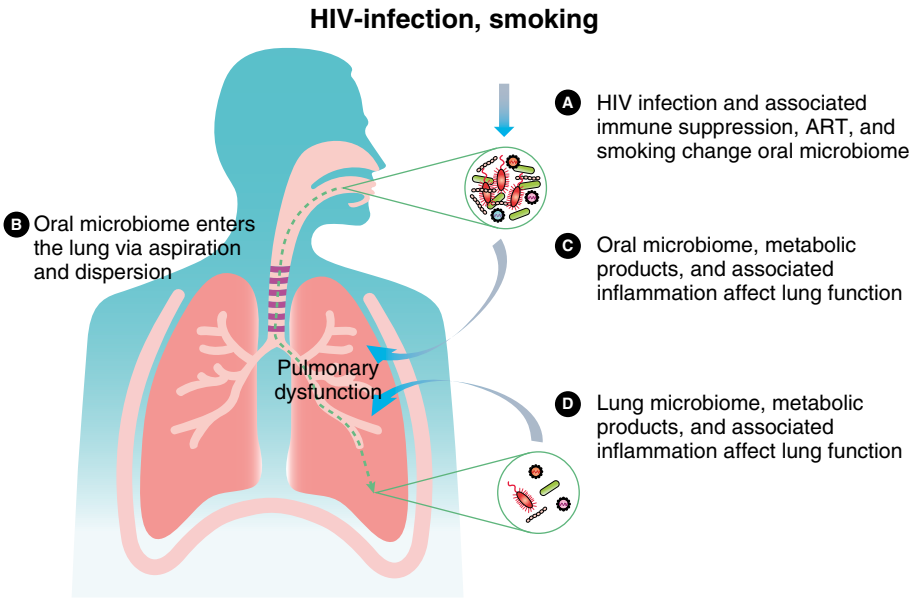


Figure 6. Role of oral microbiome in the pathogenesis of chronic obstructive pulmonary disease in persons with HIV. ART = antiretroviral therapy.

general can be altered by emphysema, pulmonary vascular disease, and other processes. In HIV, we commonly see D_{LCO} impairment, even in never smokers, suggesting that different mechanisms may contribute to its development (1, 47, 73). The oral microbiome and associated inflammation may act on different pathways to contribute to FEV_1/FVC or D_{LCO} reduction. However, it should be noted that only three of nine detected biomarkers were significantly correlated with pulmonary function and key bacteria, and these associations may have been influenced by our sample size, suggesting that microbiome may affect pulmonary function through other pathways (e.g., adaptive immunity). Alterations in the oral microbiome may be at least partially responsible for changes seen in the lung function in PWH, although it is not possible to prove causation in a cross-sectional study. Altered oral microbiota may affect local and systemic immunity and inflammation that may further contribute to lung dysfunction (74, 75). In addition, oral microbes enter the lung via subclinical microaspiration and dispersion along bronchial mucosa to directly shape the lung microbiome community and indirectly influence the growth of lung bacteria. These oral microbes and their metabolic products could stimulate inflammation, recruiting white blood cells to the lung with release of inflammatory cytokines and chemokines and proteases (47) (Figure 6).

Interestingly, these associations seemed to be unique to PWH. Among HIV-uninfected individuals, we noted that certain oral bacteria were associated with lung dysfunction in the HIV-uninfected group, but in analyses controlling for smoking and age, there was no overall difference between those with and without lung dysfunction. The interaction of HIV and lung function had a significant effect on β diversity of the oral microbiome. Possible explanations for these differences could include increased aberrant immune activation in response to oral microbes in PWH or structural differences in the oral cavity or lung environment that contribute to alterations in microbial communities or microbial function. Microaspiration may have occurred more frequently in the HIV-infected individuals and thus could result in a greater bacterial exposure in the lung, worsening the chronic inflammation responses.

In addition to providing information on pathogenesis of oral microbes in HIV and COPD, our results suggest the utility of oral sampling as a marker of disease. Lung microbiome studies have been challenging, given the difficulty and expense associated with bronchoscopy (76). Low biomass of bacteria in the lung makes sequencing difficult and increases risk of environmental contamination (77). Saliva is an easily collected sample with a higher biomass. Previous studies have shown potential overlap between oral and lung microbiome (9). Thus, the potential to use saliva as a biomarker of lung environment and lung function in PWH could greatly enhance the ability to investigate lung diseases and facilitate tracking changes over time in health and disease. Manipulation of the oral microbiome could be a novel therapeutic modality for altering lung function; prior work in the HIV-uninfected population suggests that dental interventions can improve respiratory symptoms (62).

Our study has several limitations. First, this MACS cohort may not reflect the current HIV/AIDS epidemic in the United States because it included only male participants and the majority were white; furthermore, most PWH were on appropriate ART. In addition, the study was cross-sectional and cannot determine relationships of the oral microbiome to development or progression of COPD in PWH. Second, we had limited data on oral health that was collected as part of the MACS study visit and could not investigate relationship of the oral microbiome to oral health. Moreover, the effect size (R^2) of examined variables to the microbiome was relatively small, indicating that other factors that were not included in our study might have contributed to the microbiome alterations. We also had few participants with severe COPD, which could limit differences detected in the microbiome. There may have been a smaller effect in the HIV-uninfected group that we had insufficient power to detect. Third, we did not have paired oral and lung microbiome samples. Although the similarity of lung and oral samples has been shown previously, there is not 100% overlap, and the fact that we found alterations in the oral microbiome associated with HIV, smoking, and lung function that were not reported in the lung suggest that there are important differences. These differences may occur even if the oral microbiome is largely responsible for populating the lung microbiome because BAL dilutes bacterial

communities, which may render the bacteria below the limit of detection, and aspiration of oral microbes is likely intermittent and could be missed during a single bronchoscopy. Finally, we performed 16S sequencing, and it is possible that whole-genome sequencing or metatranscriptomics might have uncovered additional differences in these populations.

Conclusions

In summary, we discovered a relationship of overall oral microbial community structure to FEV_1/FVC and D_{LCO} in PWH and reduced richness of the oral microbiome in PWH with impaired D_{LCO}. Among PWH, alterations in oral microbiome β diversity were strongly associated with both airway obstruction and diffusion impairment, and richness of the oral microbiome was reduced in PWH who had impaired diffusing capacity. *Veillonella*, *Streptococcus*, and *Lactobacillus* were increased in PWH who had abnormal lung function, suggesting that the oral microbiome may play a role in HIV COPD pathogenesis. Associations of microbes with lung function and peripheral TNF-α, MIP-1β, endothelin-1 levels suggest potential roles in stimulating inflammation and promoting lung damage. Longitudinal evaluation of the oral microbiome might serve as a noninvasive marker for risk of lung function impairment. Therapeutic manipulation of the oral microbiome to affect lung function is also an appealing, albeit untested, novel treatment modality. ■

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